Yandex Cloud Outages as a Natural Experiment on Russian Influence Operations: Bayesian Outage Modelling with Change-Point Attribution

Evidence from the October 2026 drone strikes on Yandex data centers, measured across Telegram, Reddit, the Pravda network, Russian state media and influence operations named by the EEAS

Abstract

Between 7 and 11 October 2026, drones struck Yandex data centers in Sasovo, Kaluga and Vladimir. This study systematically investigates whether the resulting outages reduced the output of Russian information operations (IO). An expected-outage function estimated from graded published reports reconstructs the measured outage onsets to within 2 and 19 minutes. The output of 243 series, covering the Pravda network, 49 Telegram channels of state media, military bloggers and foreign-audience outlets (with 16 control channels), the websites of Russian state media and of influence operations named by the EEAS, and 18 Reddit communities, is compared with out-of-sample seasonal counterfactuals using placebo windows, a Bayesian hierarchical model and 10-minute change-point timing. Only the Pravda network shows a reduction time-locked to an outage: its output fell to 64% of expected for 11 hours, starting 42 minutes after the Vladimir outage. No other platform group shows a credible drop. The strikes disrupted Yandex substantially but did not measurably silence Russian information operations; the one observed effect was narrow and transient.

October 2026

BOTTOM LINE

The strikes caused a major Yandex Cloud outage but no broad reduction in Russian information operations.

OUTAGE

Yandex Cloud lost 34% of its reachable capacity after Sasovo and 78% after Vladimir, with no recovery by 13 October.

EFFECT FOUND

The automated Pravda network stalled for 11 hours, starting 42 minutes after the Vladimir outage.

NO EFFECT

Telegram, Reddit, state media and EEAS-named outlets continued at expected levels or rose with coverage of the strikes.

Key Judgments

1 **The strikes caused a major, sustained loss of Yandex Cloud capacity.**

Almost certain (95 to 99%); high confidence.

Measured reachability of Yandex Cloud fell by 34% after the Sasovo strike (night of 7 to 8 October) and by 78% in total after the Vladimir strike (11 October), with no recovery by 13 October [1]. The Kaluga strike (9 October) left no measurable cloud signature. Operator status pages, the regional governor, NetBlocks, Cloudflare Radar and NASA FIRMS agree to within minutes [2, 3, 4, 5, 6].

2 **The outage temporarily stalled one automated influence pipeline, the Pravda network.**

Likely (55 to 80%); moderate confidence.

The network's output fell to 64% of expected for 11.2 h, starting 42 min after the measured Vladimir outage (change-point scan $p = 0.0025$), across its language editions, while the Telegram channels it republishes did not dip; 36% of the backlog was published later (Figure 1B). Confidence is moderate because Sasovo produced no comparable dip and the network's hosting, behind Cloudflare, cannot be observed [7].

3 **Russian information-operation output on the observed platforms continued at expected levels.**

Very likely (80 to 95%); moderate confidence.

Across 96 channels, language editions, communities and websites, no platform group other than the Pravda network shows a credible drop (Figure 2). This includes Telegram state media, military bloggers, tabloids and foreign-audience channels, outlets on Yandex 360 mail, and the outlets of African Initiative and RRN named by the EEAS [8]. Relative to controls, IO groups moved +8% (90% credible interval +1% to +16%) in the 24 h after Vladimir. The judgment covers public output; X, VK and closed bot infrastructure could not be observed.

Note. Likelihood follows the estimative scale of Intelligence Community Directive 203 [9]; confidence reflects the quality, independence and coverage of the evidence.

1. Introduction

Yandex operates Russia's dominant search engine and one of its largest cloud platforms. Between 7 and 11 October 2026, drones struck its data centers in Sasovo, Kaluga and Vladimir, and the resulting outages raised the question of whether Russian information operations, which depend on online infrastructure for production and distribution, were disrupted as well [10]. This study investigates that question as a natural experiment with three treated events and one internal control period, an unexplained Yandex Cloud dip on 27 September. Each link of the causal chain, from strike to capacity loss, dependent services, automated pipelines and social-media and web output, is measured separately, so that a break in the chain is visible (Section 2). The results are reported in Section 3 and discussed in Section 4.

2. Methodology

2.1 Expected Outage Function

Each strike contributes a capacity-weighted damage curve whose timing, onset lag, failure ramp, damage and recovery follow from graded reports [2, 11, 3, 6, 5, 4, 12]. Source grades (NATO Admiralty system) are converted into probabilities and pooled into each strike's credibility. The model is estimated in three nested Bayesian stages with adaptive-tempering sequential Monte Carlo [13] on a 10-minute grid: attack reports only (S0), all published information (S1), and fused with active-probing telemetry from IODA (S2) [14, 15]; BGP visibility is taken from RIPE Stat [16]. Stage 1 is validated out of sample against the measurement (Figure 1A).

2.2 Actors and Outcome Data

The actor set follows public attributions: the Pravda network (Portal Kombat) [7] and African Initiative, RRN and R-FBI [8]. It is extended with Russian state broadcasters, the Ministry of Defence broadcaster Zvezda, the Channel One Kremlin pool, the Kremlin-linked "War on Fakes" channel, foreign-audience outlets (InfoDefense, Strategic Culture, Sputnik Africa, DD Geopolitics) and the most prominent military bloggers. Output is measured as publication timestamps from Pravda sitemaps [17], Telegram channel previews [18], WordPress feeds and sitemaps [19, 20] and Reddit [21]; independent Russian, Ukrainian, business and non-political sources serve as controls. Russian Wikipedia pageviews [22] are a positive control for user-facing disruption: Yandex Search refers a substantial share of visits while Wikimedia's own infrastructure is unaffected, so a dip there shows that the outage reached ordinary users. Yandex dependency is mapped from public DNS records with network attribution from RIPE Stat.

2.3 Statistical Inference

Each series receives an out-of-sample seasonal counterfactual (hour of day, day of week and trend, fitted on the baseline only) [23]. Inference combines placebo-in-time windows of 6 and 24 hours [24] with a same-weekday check; a Bayesian hierarchical model that pools 96 member series into platform groups [25], with a Yandex-dependency term, the common news-shock variance and convergence checked by R-hat [26]; change-point dating of dips against the outage onset, calibrated with 400 placebo origins; dose-response on the expected outage function with HAC errors [27]; war-minus-non-war theme contrasts; false-discovery control [28]; a 20% equivalence bound [29]; and analysis of competing hypotheses [30]. A drop is credible only if the posterior probability of a drop larger than 10% exceeds 90%, the group placebo p is below 0.05 and the same-weekday check agrees; a dip is time-locked only if it is significant and starts at or after the onset.

2.4 Data Verification

After the analysis, random samples of every stored source (149 values) were requested again from the original services, bypassing the local cache, and compared value by value. All values match exactly for IODA (48), Wikimedia (5), Pravda network (10), Telegram (53), Web (WordPress API) (24), RIPE Stat (3); for Reddit (Arctic Shift), 83% of sampled hours match and the largest difference is 1 item, consistent with comments deleted or added after collection. Every timestamp was traced to its definition at the source: Pravda sitemap *lastmod* equals the published time, WordPress *date_gmt* is the exact publication time (WordPress sitemap dates record later edits and were not used), and Telegram previews carry each post's UTC time, reproduced to the second on re-reading. Arctic Shift labels its Reddit buckets two hours early; the offset was corrected and confirmed against the raw comment stream, and Reddit group totals are reported only when every member community is observed. A completeness audit of all 190 streams flagged 12 (5 empty, 7 sparse), which were excluded from inference.

3. Results

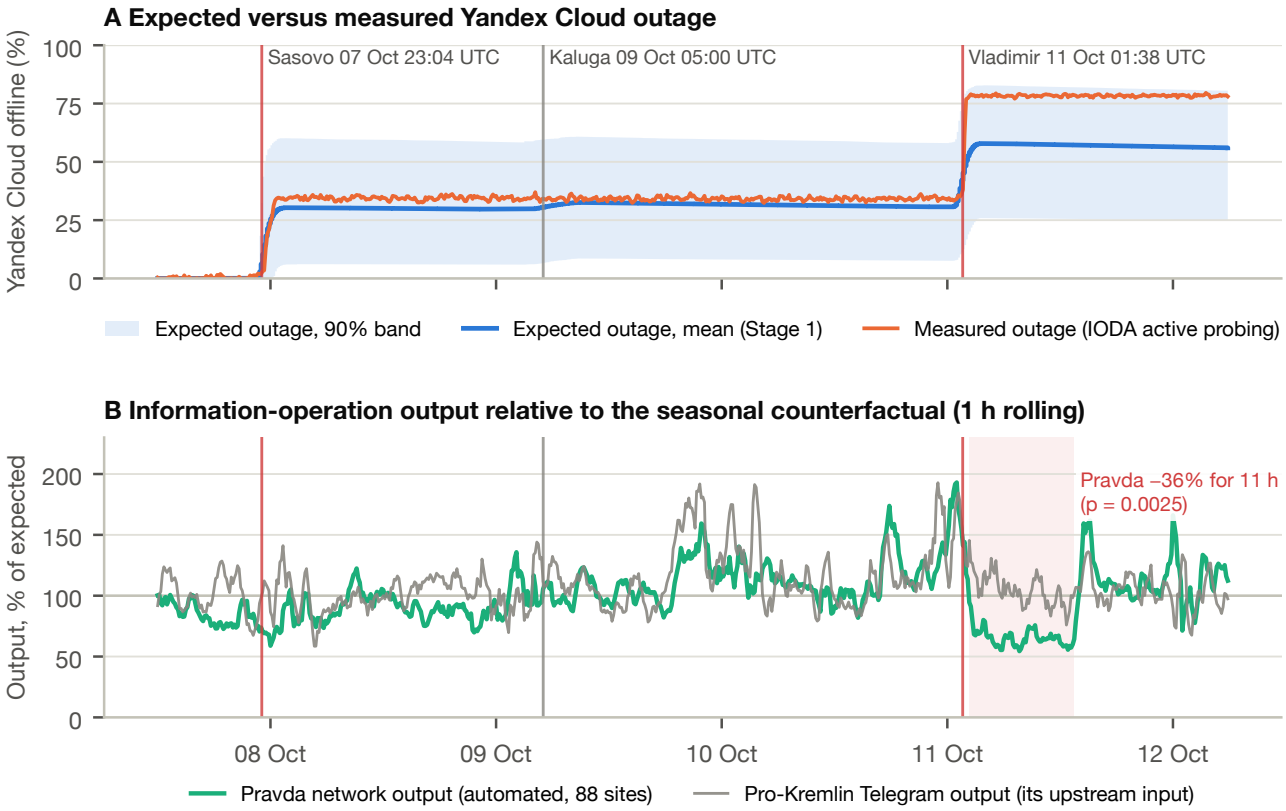
3.1 Infrastructure and Dependent Services

Yandex Cloud reachability fell from 23:04 UTC after the Sasovo strike (strike 22:27, full loss 00:24) and from 01:38 UTC after the Vladimir strike (strike 00:10) (Figure 1A). The expected outage function built from published reports alone reproduces these onsets to within 2 and 19 minutes. BGP visibility stayed intact [16], indicating a data-plane loss rather than a

routing withdrawal. Hosting, access and platform controls (Selectel, Timeweb, Beget, VK, Rostelecom, MTS, MegaFon, Beeline and Yandex Cloud Kazakhstan) show no matching steps. Russian Wikipedia user traffic fell −7% in the 24 h after Vladimir, and complaint alerts for Yandex Pay, Go and KinoPoisk surged from 00:47 UTC [12]: the outage reached consumers.

Figure 1

Expected and Measured Yandex Cloud Outage and Pravda Network Output, 7 to 12 October 2026



Note. (A) The expected outage function estimated from published reports only (statements, status pages, satellite fire detections, complaint alerts) tracks the independently measured outage [1], with 99% coverage of the 90% predictive band. (B) Pravda network output [17] collapses within the hour after the Vladimir outage and recovers with a partial catch-up; its Telegram input does not dip. No comparable dip follows Sasovo, where a decline started 7 h before the strike. Shading marks the most likely dip interval; vertical lines mark outage onsets (UTC).

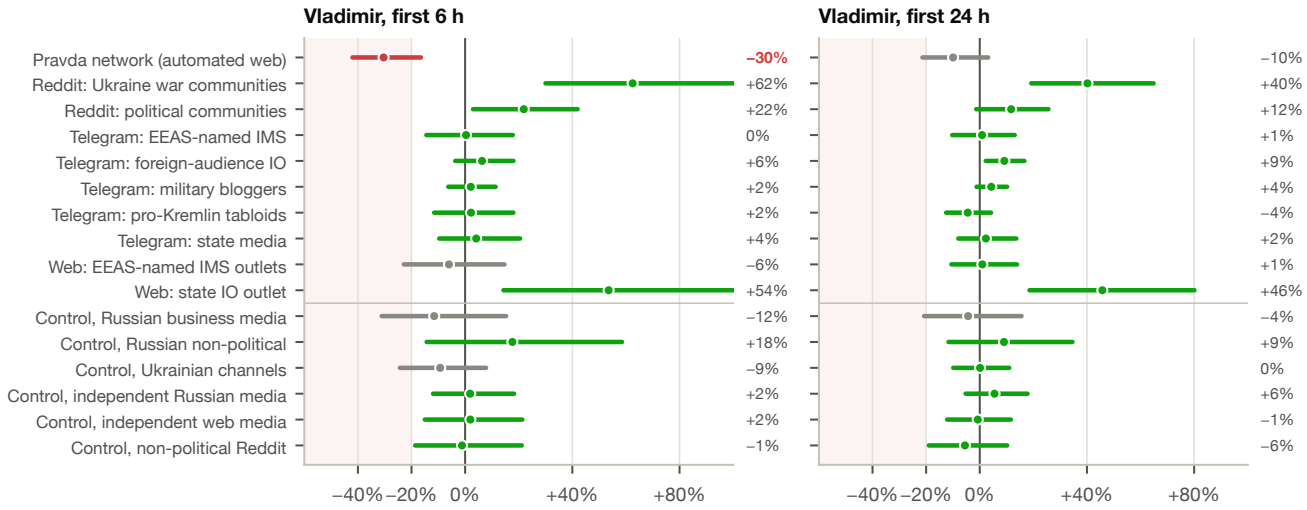
3.2 Where an Effect on Information Operations Is and Is Not Seen

- **Pravda network.** The only credible and time-locked drop: output fell to 64% of expected for 11.2 h, starting 42 min after the Vladimir onset (scan p = 0.0025; first 6 h −30%, P(drop > 10%) = 99%; Figure 1B, Figure 2). The English, German and French editions dip 22 min after onset.
- *Telegram.* State media, military bloggers, tabloids and foreign-audience channels show no credible drop. Non-war posts of state (−25%) and tabloid channels (−19%) fell while their war posts rose (+20% and +18%); this is topic substitution toward strike news, not lost capacity.
- *EEAS-named outlets.* African Initiative and RRN websites (+1% over 24 h) and African Initiative's Telegram channels (+1%) show no credible drop and no time-locked dip.
- *Yandex 360 mail.* Outlets on Yandex mail show an extra effect of −4% (90% −12% to +5%), indistinguishable from the −5% seen in the control period.
- *Web and Reddit.* The RIA outlet ukraina.ru and the pro-Russian, war and political Reddit communities show no drop; war communities rose with strike news. No war-specific reduction is found on any platform.

Of 129 IO series, 70 rule out a drop larger than 20% after Vladimir, 41 are inconclusive, 18 increased, and no individual series shows a drop that survives false-discovery control and the same-weekday check; the Pravda effect is visible at the group level and in the 10-minute timing analysis. In the analysis of competing hypotheses, a narrow dependency of specific automated pipelines carries no weighted inconsistency, whereas a broad collapse of IO output is contradicted by every platform-level measurement.

Figure 2

Bayesian Posterior Change in Output per Platform Group After the Vladimir Strike



Note. Posterior median (value at right) and 90% credible interval, including the common news-shock variance estimated before the strikes. Red: credible drop. Green: drops larger than 20% ruled out. Grey: inconclusive. The shaded band marks drops beyond the 20% smallest effect of interest.

4. Conclusion

The October 2026 drone strikes by the Ukrainian armed forces on Yandex data centers produced a large and sustained outage of Yandex Cloud, and published reports alone were sufficient to reconstruct its timing to within minutes. The evidence does not support the view that the strikes silenced Russian information operations. Their only measurable effect on IO output was an 11-hour stall of the Pravda network after the Vladimir strike, a pro-Kremlin amplification network that automatically republishes state media and Telegram content rather than producing original disinformation, so the stall reduced the reach of existing narratives rather than their production. This pattern is consistent with a dependency of one republishing pipeline on lost capacity rather than with a broad loss of capability. The human-operated Telegram channels of state media, military bloggers and foreign-audience outlets continued and partly intensified with coverage of the strikes.

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